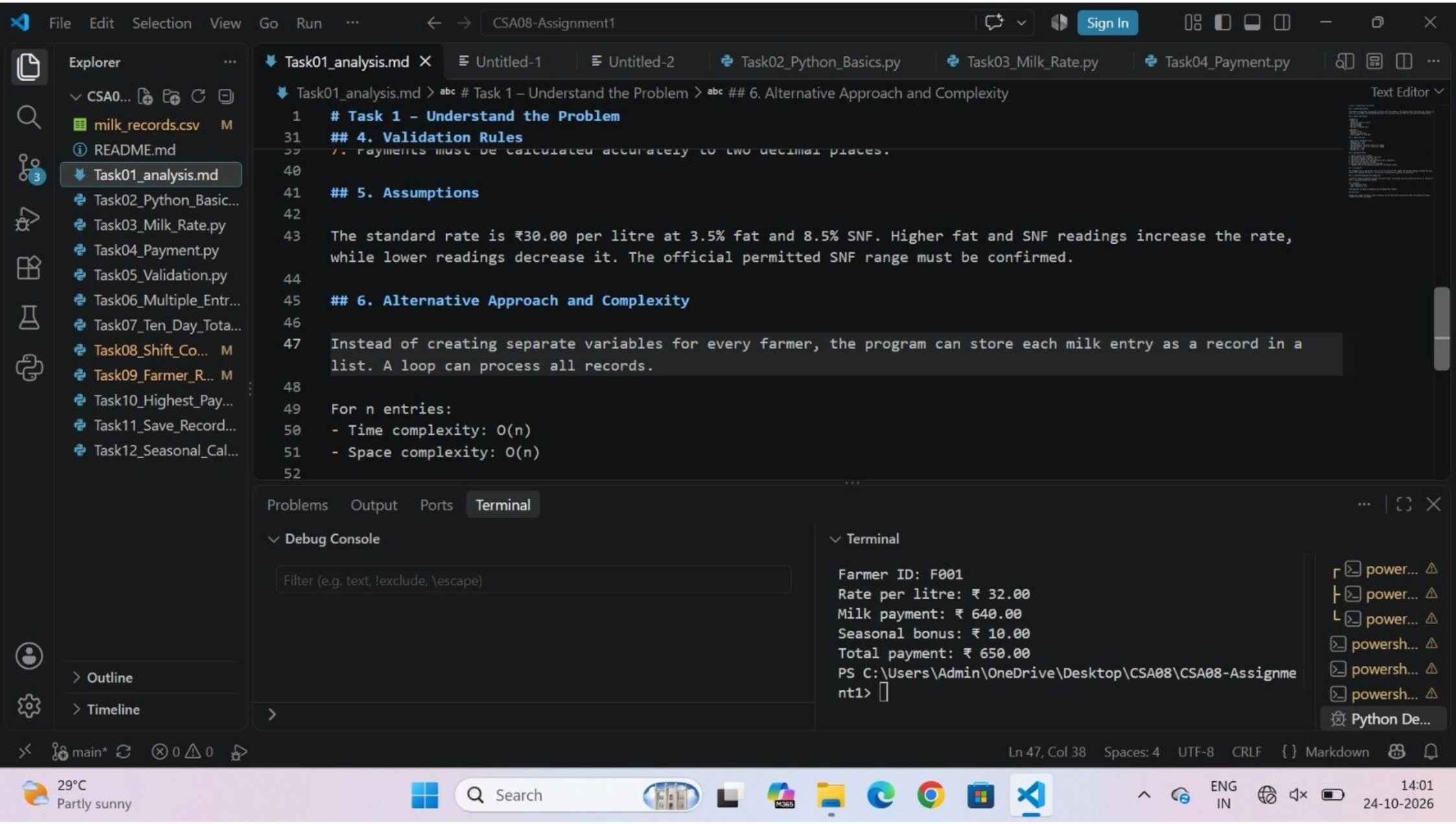
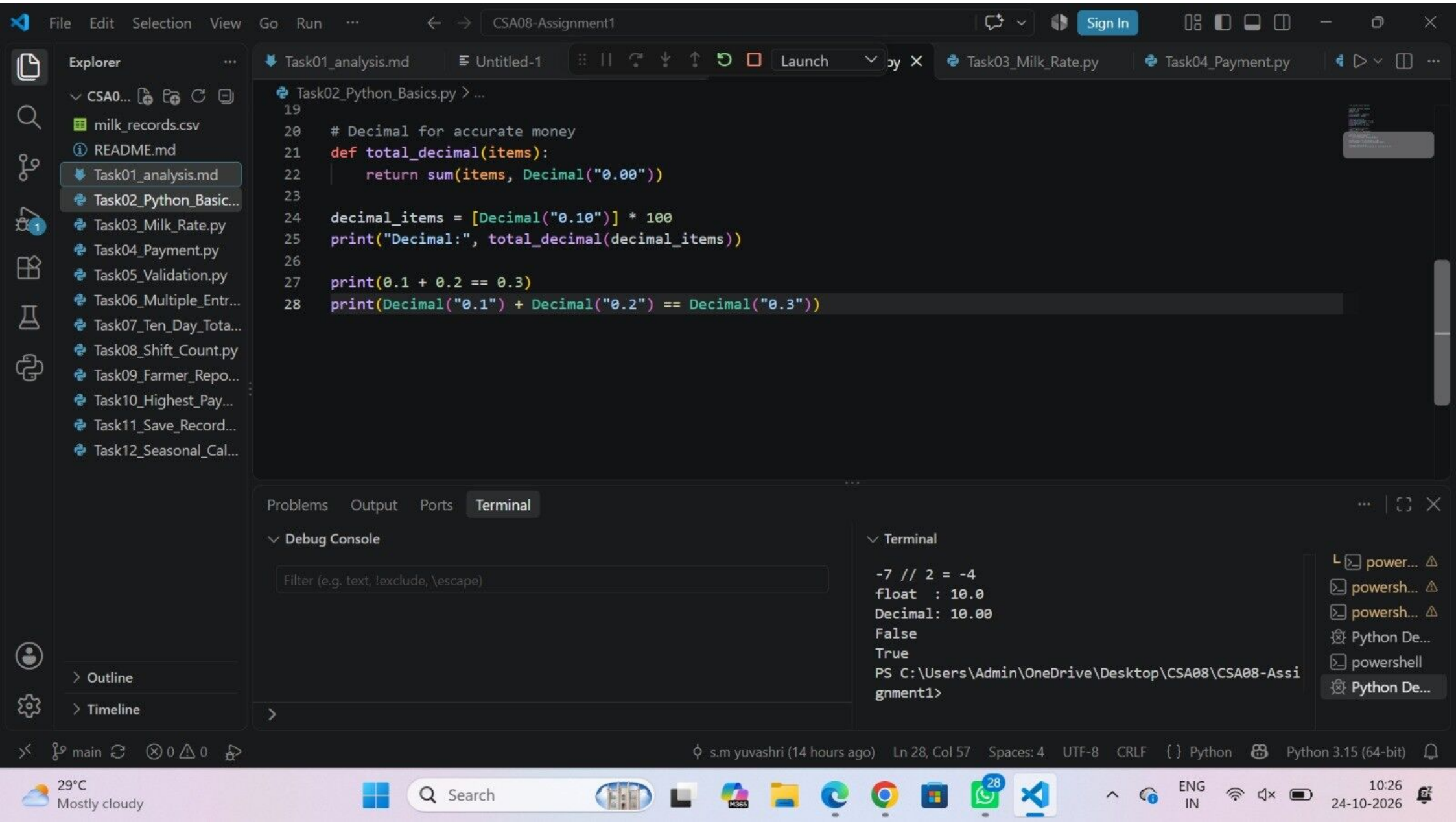
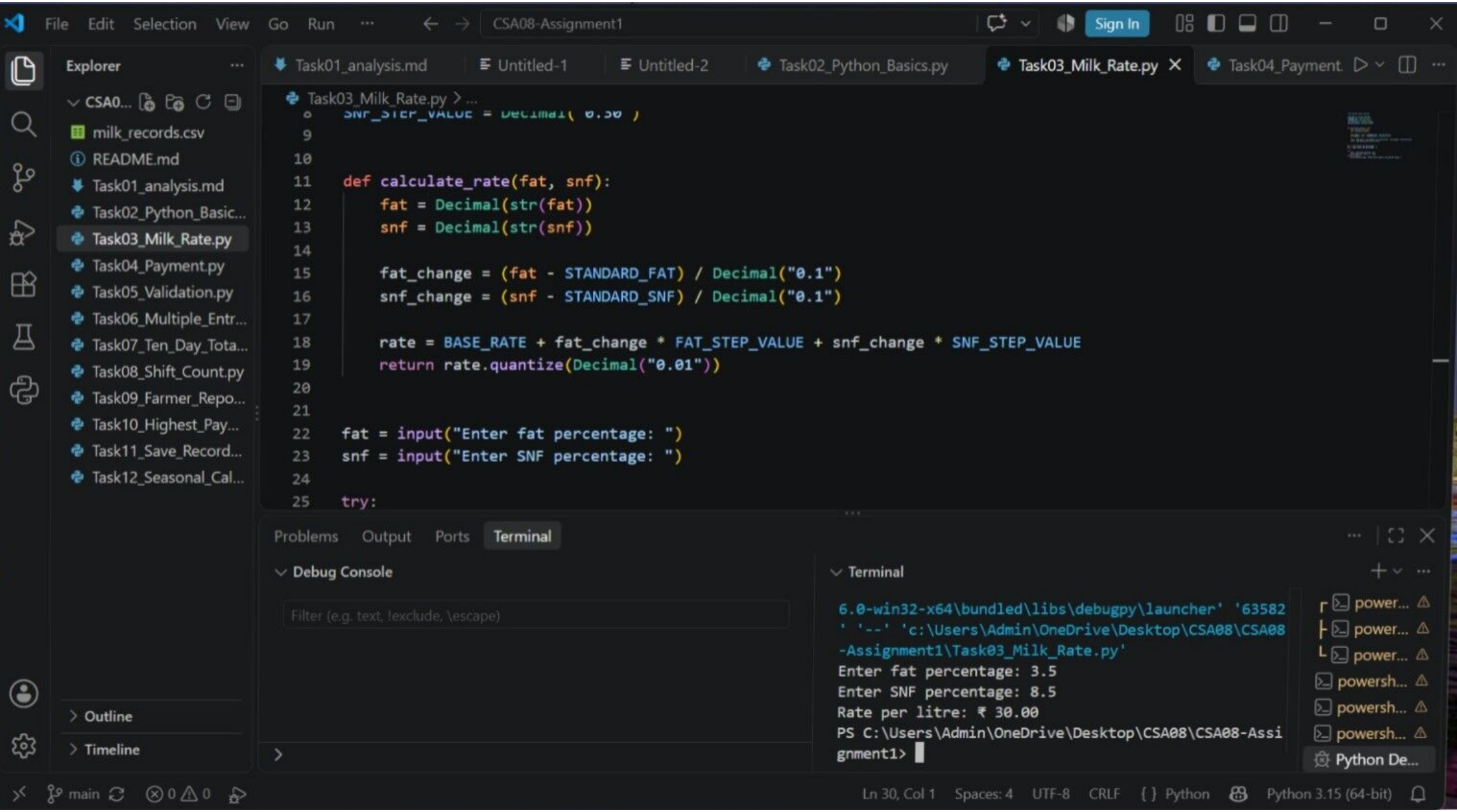
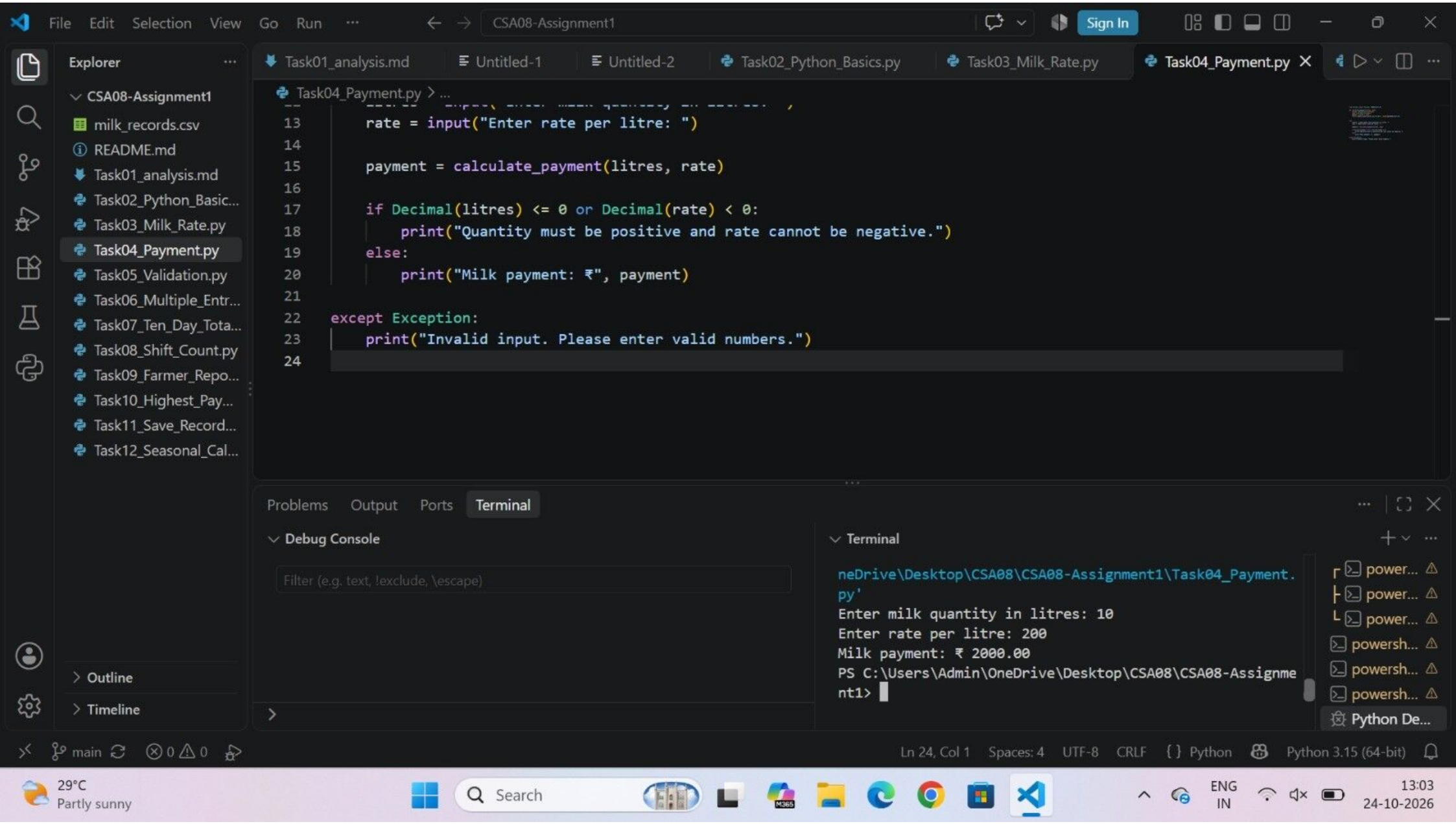
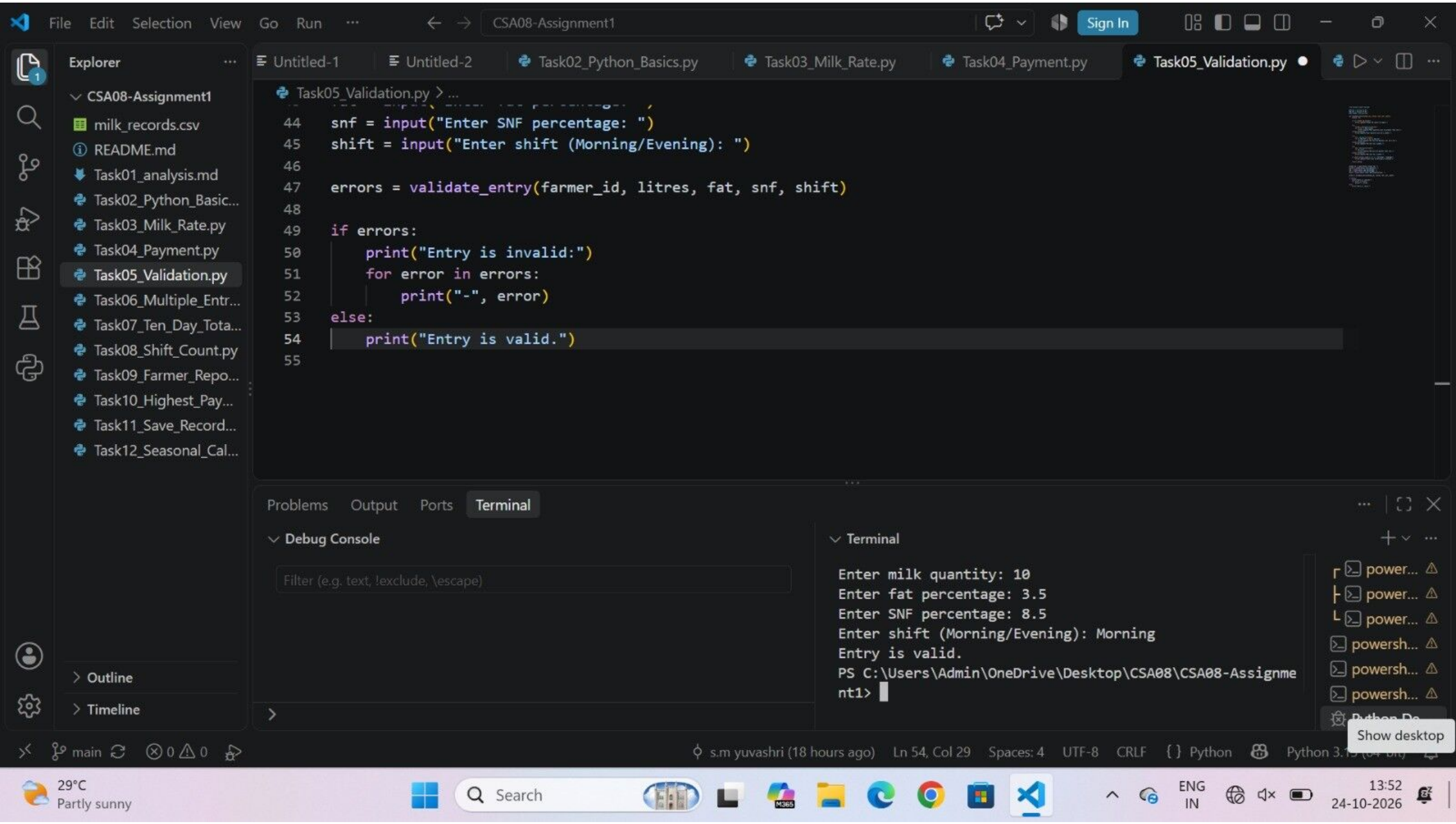
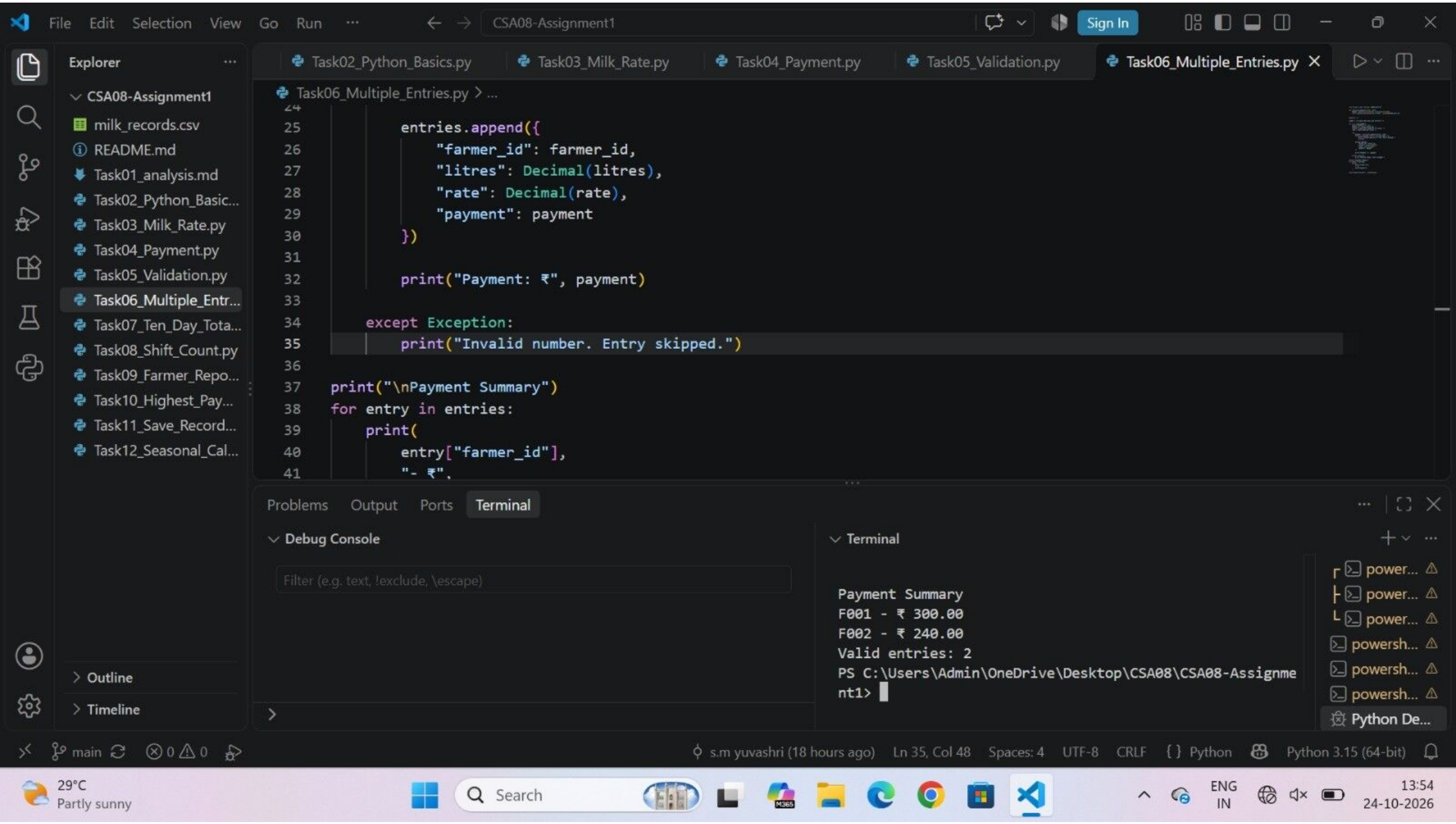

File Edit Selection View Go Run
CSA08-Assignment1
Sign In
Explorer
CSA0...
milk_records.csv M
README.md
Task01_analysis.md
Task02_Python_Basic...
Task03_Milk_Rate.py
Task04_Payment.py
Task05_Validation.py
Task06_Multiple_Entr...
Task07_Ten_Day_Tota...
Task08_Shift_Co... M
Task09_Farmer_R... M
Task10_Highest_Pay...
Task11_Save_Record...
Task12_Seasonal_Cal...
Outline
Timeline
Task01_analysis.md
Untitled-1
Untitled-2
Task02_Python_Basics.py
Task03_Milk_Rate.py
Task04_Payment.py
Task01_analysis.md > # Task 1 – Understand the Problem > ## 6. Alternative Approach and Complexity
Text Editor
1 # Task 1 – Understand the Problem
31 ## 4. Validation Rules
7. Payments must be calculated accurately to two decimal places.
40
41 ## 5. Assumptions
42
43 The standard rate is ₹30.00 per litre at 3.5% fat and 8.5% SNF. Higher fat and SNF readings increase the rate, while lower readings decrease it. The official permitted SNF range must be confirmed.
44
45 ## 6. Alternative Approach and Complexity
46
47 Instead of creating separate variables for every farmer, the program can store each milk entry as a record in a list. A loop can process all records.
48
49 For n entries:
50 - Time complexity: O(n)
51 - Space complexity: O(n)
52
Problems Output Ports Terminal
Debug Console
Filter (e.g. text, !exclude, \escape)
Terminal
Farmer ID: F001
Rate per litre: ₹ 32.00
Milk payment: ₹ 640.00
Seasonal bonus: ₹ 10.00
Total payment: ₹ 650.00
PS C:\Users\Admin\OneDrive\Desktop\CSA08\CSA08-Assignment1>
power...
power...
power...
powersh...
powersh...
powersh...
Python De...
main*
Ln 47, Col 38 Spaces: 4 UTF-8 CRLF { } Markdown
29°C
Partly sunny
Search
ENG
IN
14:01
24-10-2026

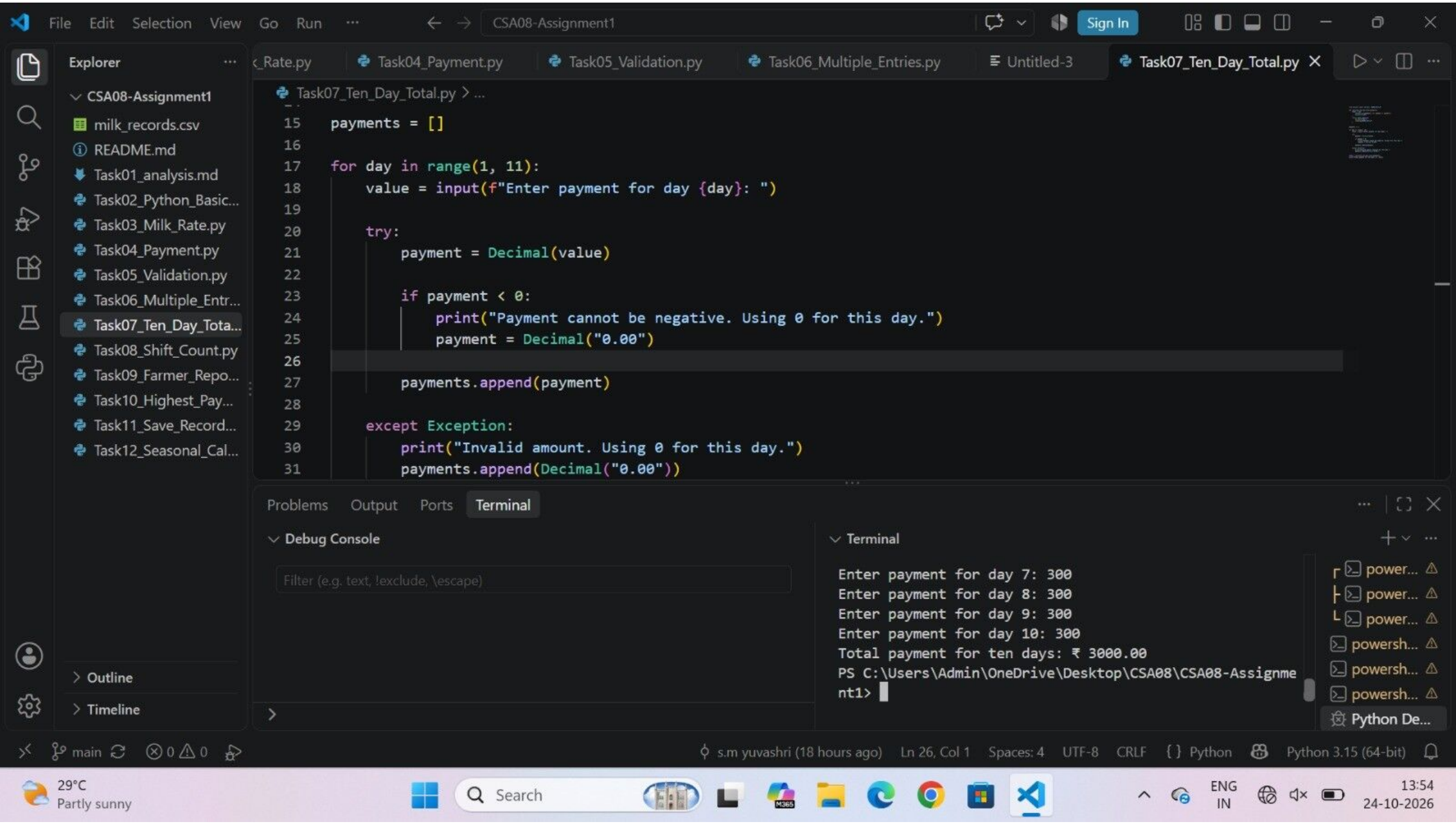

File Edit Selection View Go Run
CSA08-Assignment1
Sign In
Explorer
CSA0...
milk_records.csv
README.md
Task01_analysis.md
Task02_Python_Basic...
Task03_Milk_Rate.py
Task04_Payment.py
Task05_Validation.py
Task06_Multiple_Entr...
Task07_Ten_Day_Tota...
Task08_Shift_Count.py
Task09_Farmer_Repo...
Task10_Highest_Pay...
Task11_Save_Record...
Task12_Seasonal_Cal...
Outline
Timeline
Task01_analysis.md
Untitled-1
Launch
Task03_Milk_Rate.py
Task04_Payment.py
Task02_Python_Basics.py > ...
19
20 # Decimal for accurate money
21 def total_decimal(items):
22 return sum(items, Decimal("0.00"))
23
24 decimal_items = [Decimal("0.10")] * 100
25 print("Decimal:", total_decimal(decimal_items))
26
27 print(0.1 + 0.2 == 0.3)
28 print(Decimal("0.1") + Decimal("0.2") == Decimal("0.3"))
Problems
Output
Ports
Terminal
Debug Console
Filter (e.g. text, !exclude, \escape)
Terminal
-7 // 2 = -4
float : 10.0
Decimal: 10.00
False
True
PS C:\Users\Admin\OneDrive\Desktop\CSA08\CSA08-Assignment1>
power...
powersh...
powersh...
Python De...
powershell
Python De...
main
0 0
s.m yuvashri (14 hours ago)
Ln 28, Col 57
Spaces: 4
UTF-8
CRLF
Python
Python 3.15 (64-bit)
29°C
Mostly cloudy
Search
ENG
IN
10:26
24-10-2026

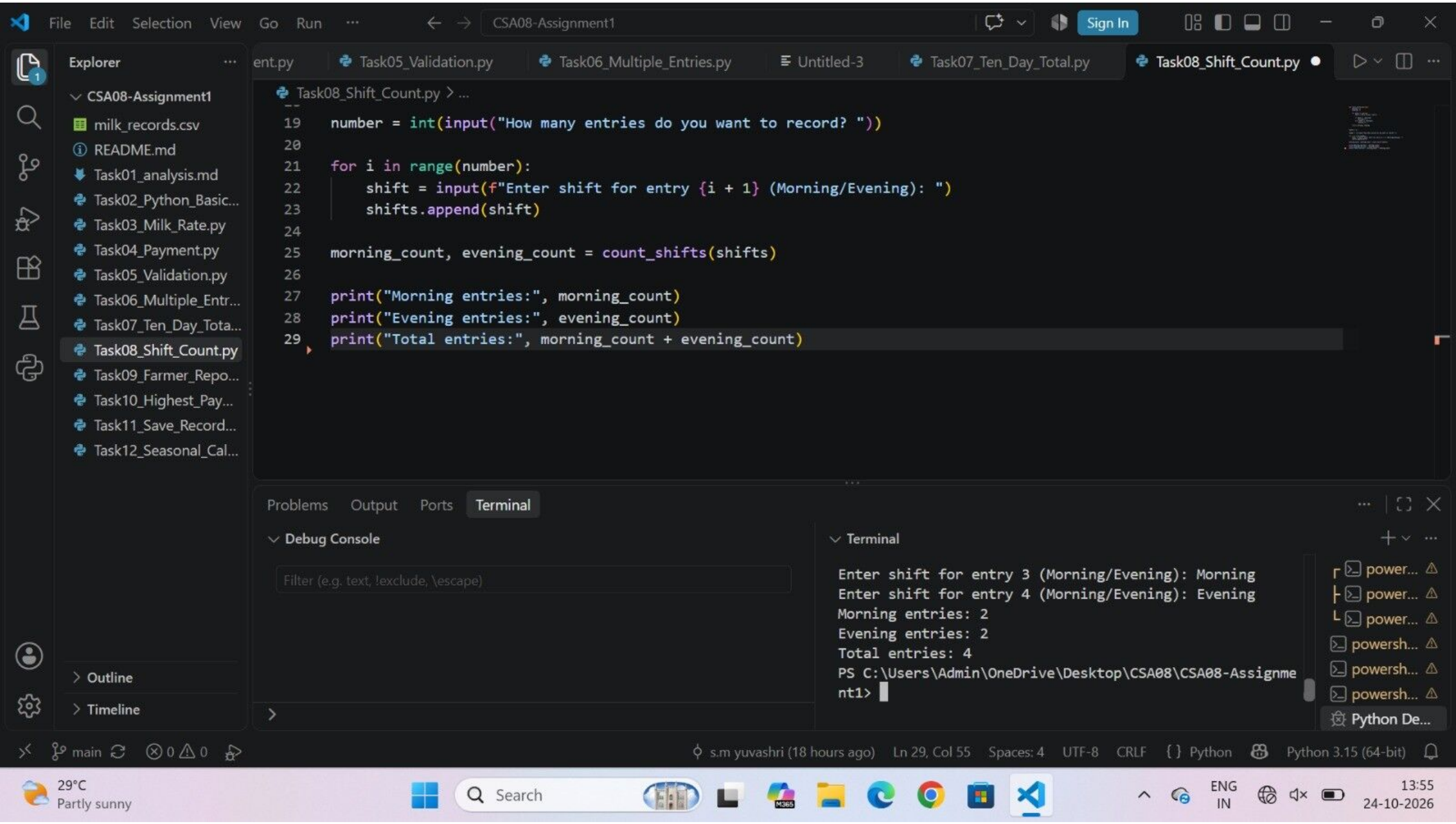

File Edit Selection View Go Run
CSA08-Assignment1
Sign In
Explorer
CSA0...
milk_records.csv
README.md
Task01_analysis.md
Task02_Python_Basic...
Task03_Milk_Rate.py
Task04_Payment.py
Task05_Validation.py
Task06_Multiple_Entr...
Task07_Ten_Day_Tota...
Task08_Shift_Count.py
Task09_Farmer_Repo...
Task10_Highest_Pay...
Task11_Save_Record...
Task12_Seasonal_Cal...
Task01_analysis.md
Untitled-1
Untitled-2
Task02_Python_Basics.py
Task03_Milk_Rate.py
Task04_Payment
Task03_Milk_Rate.py > ...
SNF_STEP_VALUE = Decimal( 0.50 )
def calculate_rate(fat, snf):
fat = Decimal(str(fat))
snf = Decimal(str(snf))
fat_change = (fat - STANDARD_FAT) / Decimal("0.1")
snf_change = (snf - STANDARD_SNF) / Decimal("0.1")
rate = BASE_RATE + fat_change * FAT_STEP_VALUE + snf_change * SNF_STEP_VALUE
return rate.quantize(Decimal("0.01"))
fat = input("Enter fat percentage: ")
snf = input("Enter SNF percentage: ")
try:
Problems
Output
Ports
Terminal
Debug Console
Filter (e.g. text, !exclude, \escape)
Terminal
6.0-win32-x64\bundled\libs\debugpy\launcher' '63582
' '--' 'c:\Users\Admin\OneDrive\Desktop\CSA08\CSA08
-Assignment1\Task03_Milk_Rate.py'
Enter fat percentage: 3.5
Enter SNF percentage: 8.5
Rate per litre: ₹ 30.00
PS C:\Users\Admin\OneDrive\Desktop\CSA08\CSA08-Assi
gnment1>
power...
power...
power...
powersh...
powersh...
powersh...
Python De...
Outline
Timeline
main
0 0
Ln 30, Col 1
Spaces: 4
UTF-8
CRLF
{ } Python
Python 3.15 (64-bit)

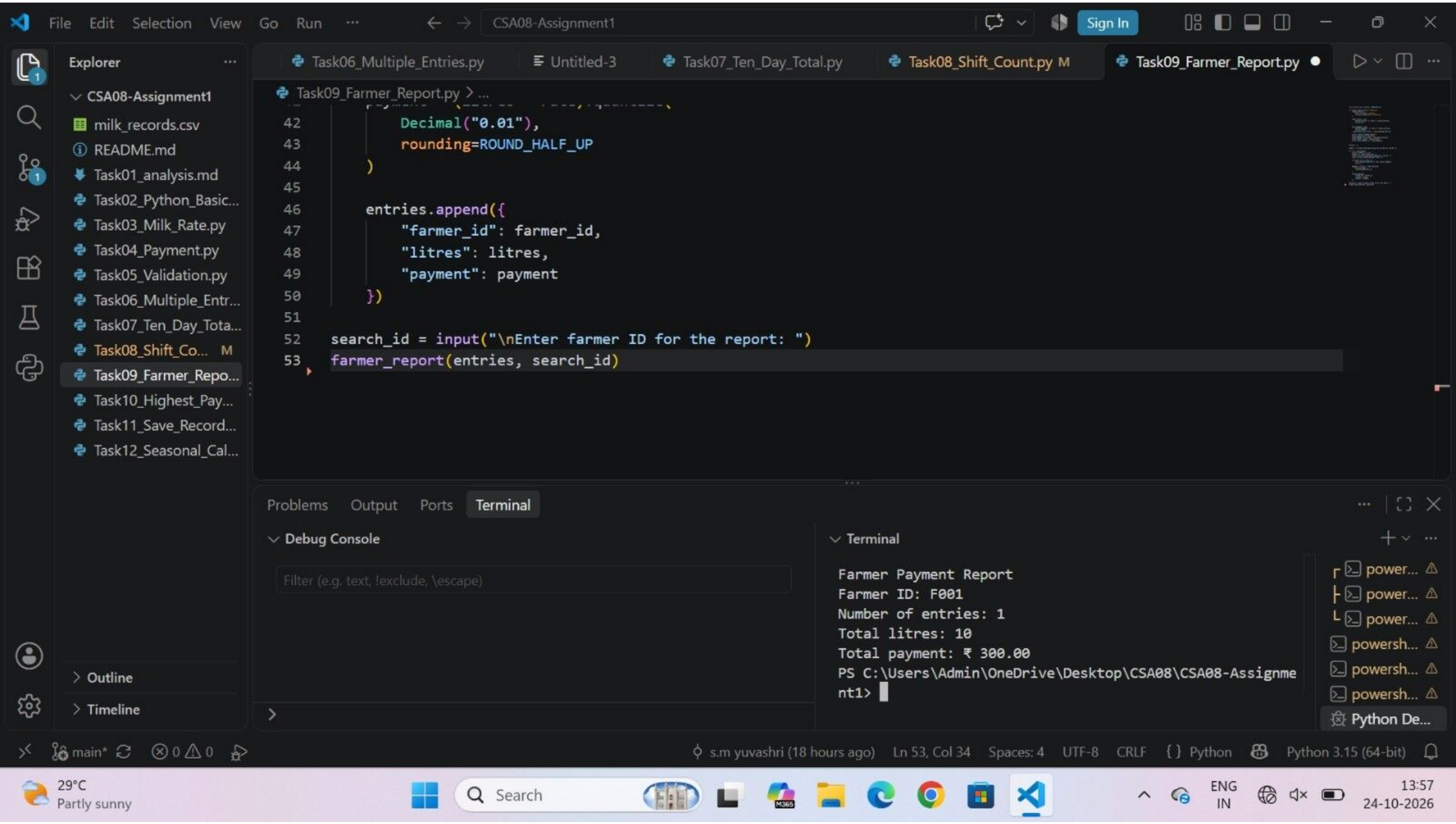

File Edit Selection View Go Run
CSA08-Assignment1
Sign In
Explorer
CSA08-Assignment1
milk_records.csv
README.md
Task01_analysis.md
Task02_Python_Basic...
Task03_Milk_Rate.py
Task04_Payment.py
Task05_Validation.py
Task06_Multiple_Entr...
Task07_Ten_Day_Tota...
Task08_Shift_Count.py
Task09_Farmer_Repo...
Task10_Highest_Pay...
Task11_Save_Record...
Task12_Seasonal_Cal...
Task01_analysis.md
Untitled-1
Untitled-2
Task02_Python_Basics.py
Task03_Milk_Rate.py
Task04_Payment.py
Task04_Payment.py > ...
rate = input("Enter rate per litre: ")
payment = calculate_payment(litres, rate)
if Decimal(litres) <= 0 or Decimal(rate) < 0:
print("Quantity must be positive and rate cannot be negative.")
else:
print("Milk payment: ₹", payment)
except Exception:
print("Invalid input. Please enter valid numbers.")
Problems
Output
Ports
Terminal
Debug Console
Filter (e.g. text, !exclude, \escape)
Terminal
neDrive\Desktop\CSA08\CSA08-Assignment1\Task04_Payment.py'
Enter milk quantity in litres: 10
Enter rate per litre: 200
Milk payment: ₹ 2000.00
PS C:\Users\Admin\OneDrive\Desktop\CSA08\CSA08-Assignment1>
Outline
Timeline
main
Ln 24, Col 1
Spaces: 4
UTF-8
CRLF
Python
Python 3.15 (64-bit)
29°C
Partly sunny
Search
ENG
IN
13:03
24-10-2026

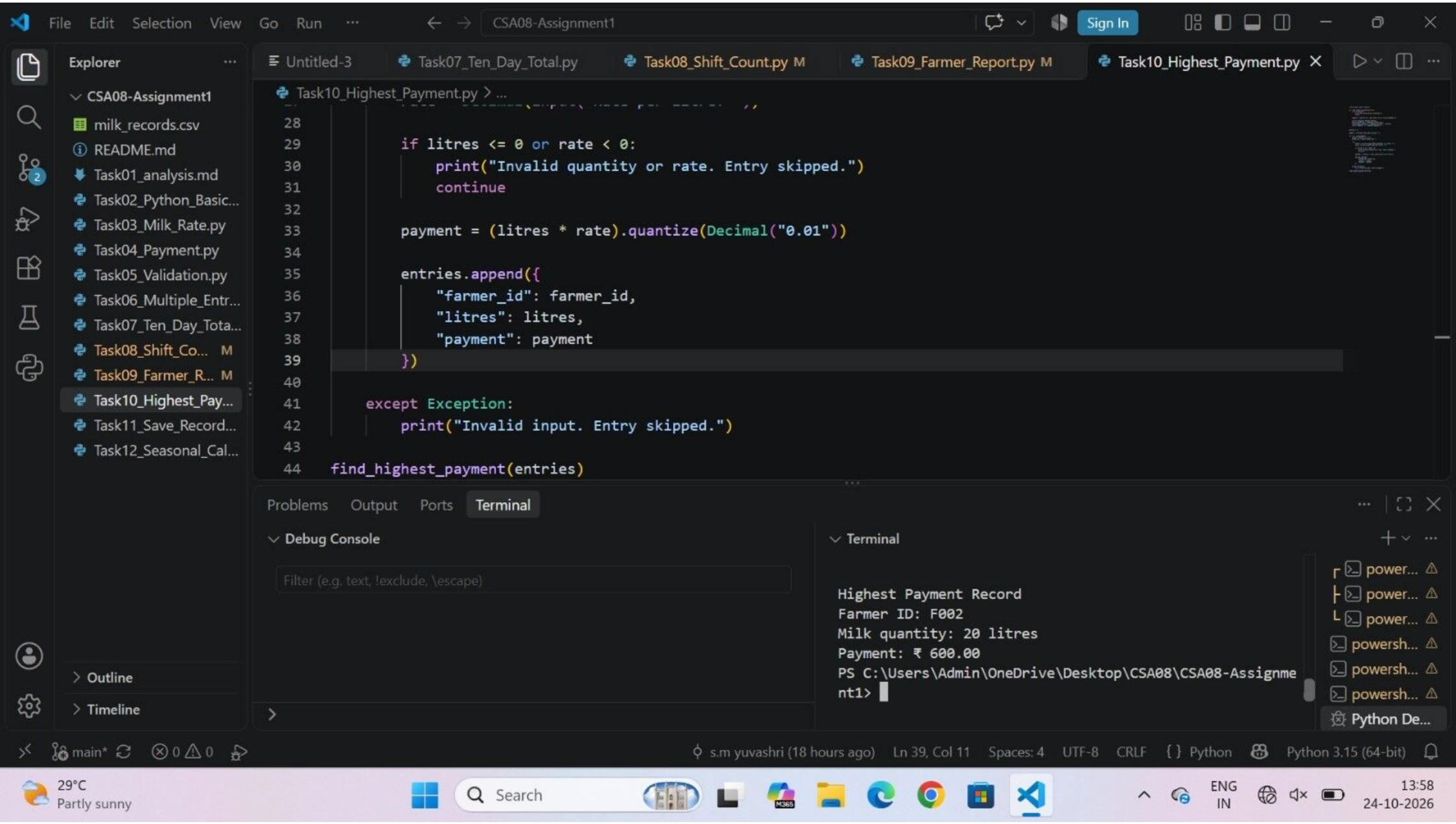

File Edit Selection View Go Run
CSA08-Assignment1
Sign In
Explorer
CSA08-Assignment1
milk_records.csv
README.md
Task01_analysis.md
Task02_Python_Basic...
Task03_Milk_Rate.py
Task04_Payment.py
Task05_Validation.py
Task06_Multiple_Entr...
Task07_Ten_Day_Tota...
Task08_Shift_Count.py
Task09_Farmer_Repo...
Task10_Highest_Pay...
Task11_Save_Record...
Task12_Seasonal_Cal...
Outline
Timeline
Untitled-1
Untitled-2
Task02_Python_Basics.py
Task03_Milk_Rate.py
Task04_Payment.py
Task05_Validation.py
Task05_Validation.py > ...
44 snf = input("Enter SNF percentage: ")
45 shift = input("Enter shift (Morning/Evening): ")
46
47 errors = validate_entry(farmer_id, litres, fat, snf, shift)
48
49 if errors:
50 print("Entry is invalid:")
51 for error in errors:
52 print("-", error)
53 else:
54 print("Entry is valid.")
55
Problems Output Ports Terminal
Debug Console
Filter (e.g. text, !exclude, \escape)
Terminal
Enter milk quantity: 10
Enter fat percentage: 3.5
Enter SNF percentage: 8.5
Enter shift (Morning/Evening): Morning
Entry is valid.
PS C:\Users\Admin\OneDrive\Desktop\CSA08\CSA08-Assignment1>
power...
power...
power...
powersh...
powersh...
powersh...
main
s.m yuvashri (18 hours ago)
Ln 54, Col 29
Spaces: 4
UTF-8
CRLF
Python
Show desktop
29°C
Partly sunny
Search
ENG
IN
13:52
24-10-2026

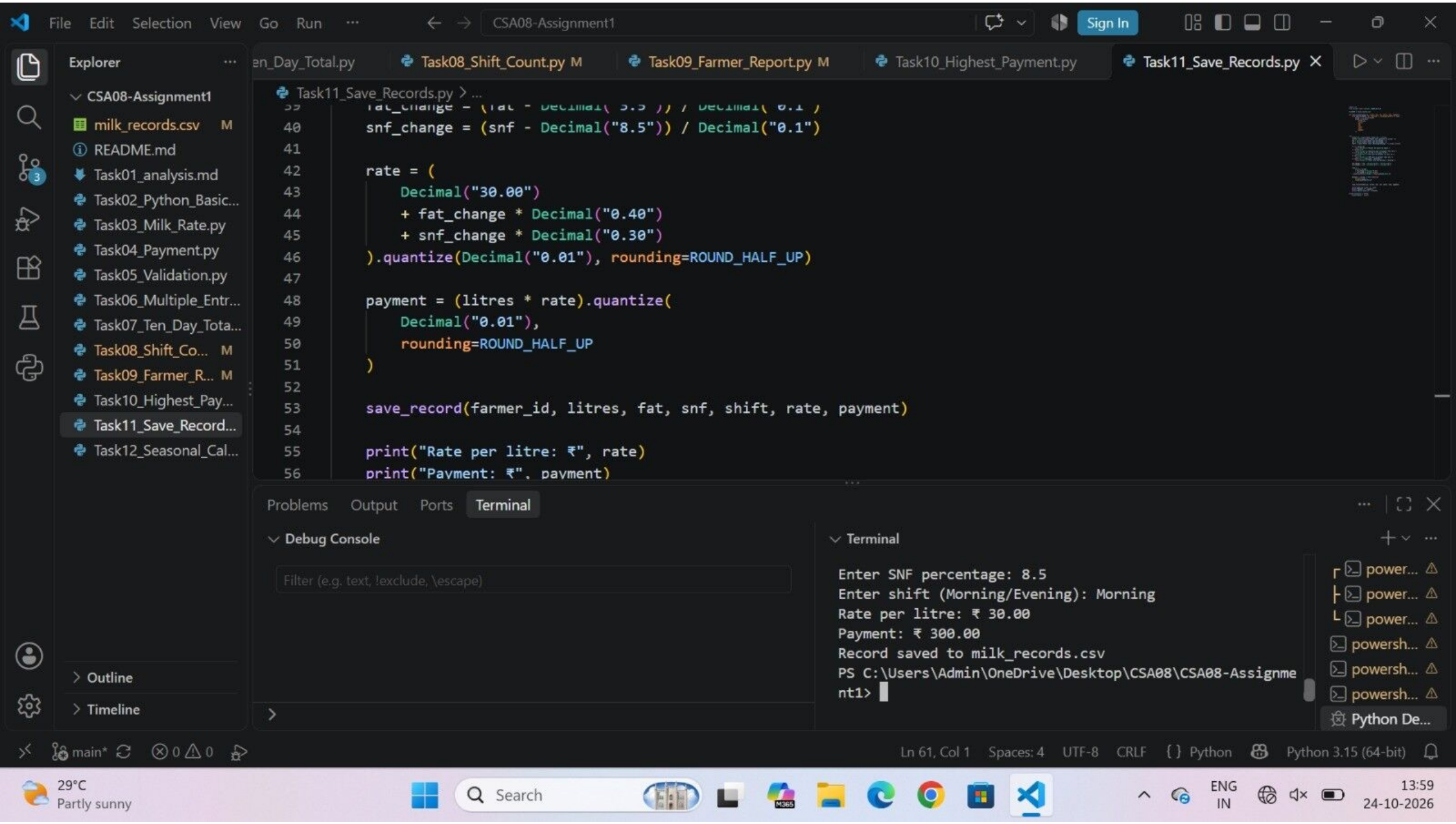

File Edit Selection View Go Run
CSA08-Assignment1
Sign In
Explorer
CSA08-Assignment1
milk_records.csv
README.md
Task01_analysis.md
Task02_Python_Basic...
Task03_Milk_Rate.py
Task04_Payment.py
Task05_Validation.py
Task06_Multiple_Entr...
Task07_Ten_Day_Tota...
Task08_Shift_Count.py
Task09_Farmer_Repo...
Task10_Highest_Pay...
Task11_Save_Record...
Task12_Seasonal_Cal...
Task02_Python_Basics.py
Task03_Milk_Rate.py
Task04_Payment.py
Task05_Validation.py
Task06_Multiple_Entries.py
Task06_Multiple_Entries.py > ...
entries.append({
"farmer_id": farmer_id,
"litres": Decimal(litres),
"rate": Decimal(rate),
"payment": payment
})
print("Payment: ₹", payment)
except Exception:
print("Invalid number. Entry skipped.")
print("\nPayment Summary")
for entry in entries:
print(
entry["farmer_id"],
Problems Output Ports Terminal
Debug Console
Filter (e.g. text, !exclude, \escape)
Terminal
Payment Summary
F001 - ₹ 300.00
F002 - ₹ 240.00
Valid entries: 2
PS C:\Users\Admin\OneDrive\Desktop\CSA08\CSA08-Assignment1>
Outline
Timeline
main
s.m yuvashri (18 hours ago)
Ln 35, Col 48
Spaces: 4
UTF-8
CRLF
Python
Python 3.15 (64-bit)
29°C
Partly sunny
Search
ENG
IN
13:54
24-10-2026

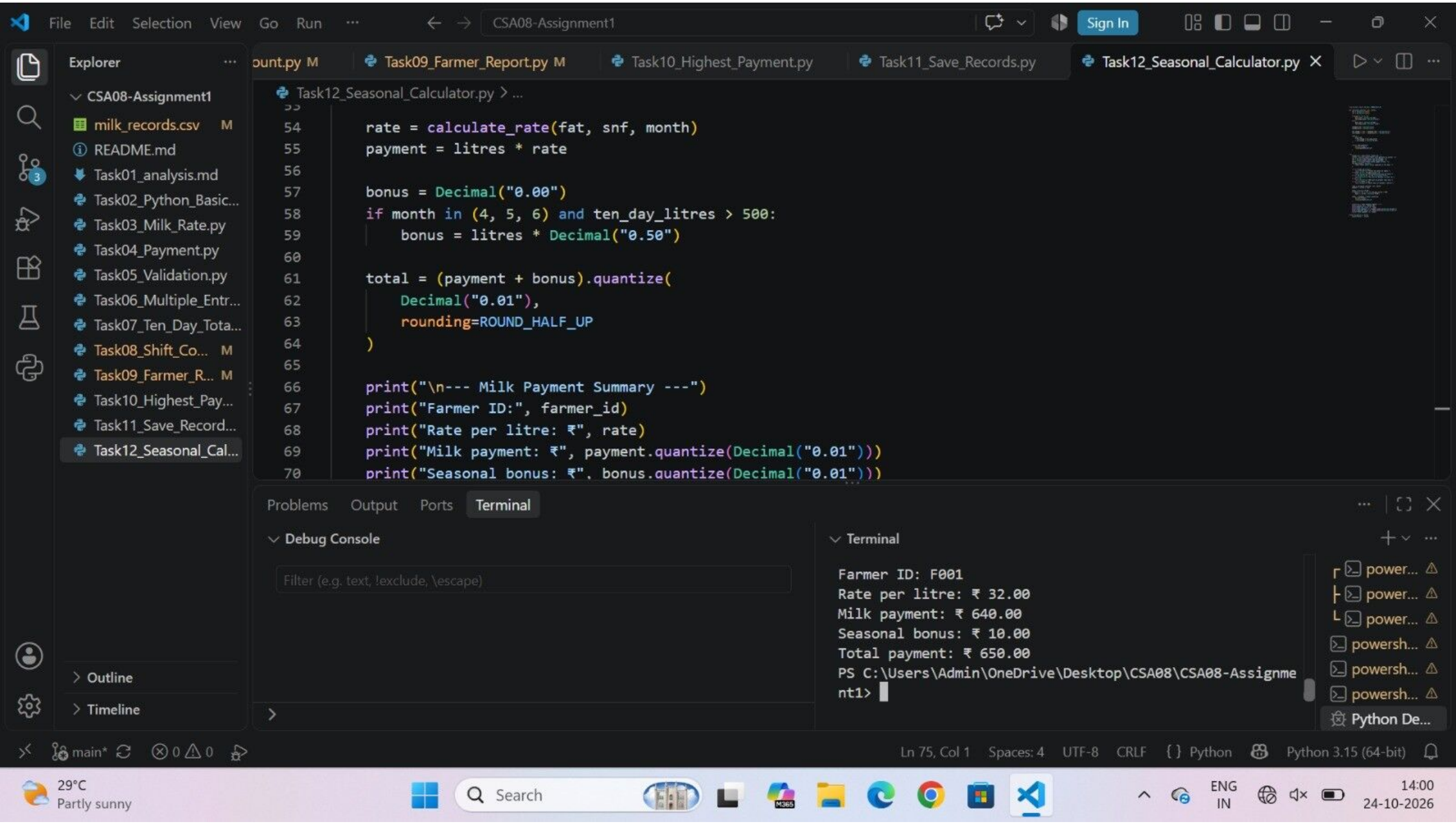

File Edit Selection View Go Run
CSA08-Assignment1
Sign In
Explorer
CSA08-Assignment1
milk_records.csv
README.md
Task01_analysis.md
Task02_Python_Basic...
Task03_Milk_Rate.py
Task04_Payment.py
Task05_Validation.py
Task06_Multiple_Entr...
Task07_Ten_Day_Tota...
Task08_Shift_Count.py
Task09_Farmer_Repo...
Task10_Highest_Pay...
Task11_Save_Record...
Task12_Seasonal_Cal...
Outline
Timeline
_Rate.py
Task04_Payment.py
Task05_Validation.py
Task06_Multiple_Entries.py
Untitled-3
Task07_Ten_Day_Total.py
Task07_Ten_Day_Total.py > ...
15 payments = []
16
17 for day in range(1, 11):
18 value = input(f"Enter payment for day {day}: ")
19
20 try:
21 payment = Decimal(value)
22
23 if payment < 0:
24 print("Payment cannot be negative. Using 0 for this day.")
25 payment = Decimal("0.00")
26
27 payments.append(payment)
28
29 except Exception:
30 print("Invalid amount. Using 0 for this day.")
31 payments.append(Decimal("0.00"))
Problems Output Ports Terminal
Debug Console
Filter (e.g. text, !exclude, \escape)
Terminal
Enter payment for day 7: 300
Enter payment for day 8: 300
Enter payment for day 9: 300
Enter payment for day 10: 300
Total payment for ten days: ₹ 3000.00
PS C:\Users\Admin\OneDrive\Desktop\CSA08\CSA08-Assignment1>
power...
power...
power...
powersh...
powersh...
powersh...
Python De...
main
0 0
s.m yuvashri (18 hours ago)
Ln 26, Col 1
Spaces: 4
UTF-8
CRLF
Python
Python 3.15 (64-bit)
29°C
Partly sunny
Search
ENG
IN
13:54
24-10-2026


File Edit Selection View Go Run
CSA08-Assignment1
Sign In
Explorer
CSA08-Assignment1
milk_records.csv
README.md
Task01_analysis.md
Task02_Python_Basic...
Task03_Milk_Rate.py
Task04_Payment.py
Task05_Validation.py
Task06_Multiple_Entr...
Task07_Ten_Day_Tota...
Task08_Shift_Count.py
Task09_Farmer_Repo...
Task10_Highest_Pay...
Task11_Save_Record...
Task12_Seasonal_Cal...
ent.py
Task05_Validation.py
Task06_Multiple_Entries.py
Untitled-3
Task07_Ten_Day_Total.py
Task08_Shift_Count.py
Task08_Shift_Count.py > ...
19 number = int(input("How many entries do you want to record? "))
20
21 for i in range(number):
22 shift = input(f"Enter shift for entry {i + 1} (Morning/Evening): ")
23 shifts.append(shift)
24
25 morning_count, evening_count = count_shifts(shifts)
26
27 print("Morning entries:", morning_count)
28 print("Evening entries:", evening_count)
29 print("Total entries:", morning_count + evening_count)
Problems Output Ports Terminal
Debug Console
Filter (e.g. text, !exclude, \escape)
Terminal
Enter shift for entry 3 (Morning/Evening): Morning
Enter shift for entry 4 (Morning/Evening): Evening
Morning entries: 2
Evening entries: 2
Total entries: 4
PS C:\Users\Admin\OneDrive\Desktop\CSA08\CSA08-Assignment1>
power...
power...
power...
powersh...
powersh...
powersh...
Python De...
Outline
Timeline
main
s.m yuvashri (18 hours ago)
Ln 29, Col 55
Spaces: 4
UTF-8
CRLF
Python
Python 3.15 (64-bit)
29°C
Partly sunny
Search
ENG
IN
13:55
24-10-2026


File Edit Selection View Go Run
CSA08-Assignment1
Sign In
Explorer
CSA08-Assignment1
milk_records.csv
README.md
Task01_analysis.md
Task02_Python_Basic...
Task03_Milk_Rate.py
Task04_Payment.py
Task05_Validation.py
Task06_Multiple_Entr...
Task07_Ten_Day_Tota...
Task08_Shift_Co... M
Task09_Farmer_Repo...
Task10_Highest_Pay...
Task11_Save_Record...
Task12_Seasonal_Cal...
Task06_Multiple_Entries.py
Untitled-3
Task07_Ten_Day_Total.py
Task08_Shift_Count.py M
Task09_Farmer_Report.py
Task09_Farmer_Report.py > ...
Decimal("0.01"),
rounding=ROUND_HALF_UP
)
entries.append({
"farmer_id": farmer_id,
"litres": litres,
"payment": payment
})
search_id = input("\nEnter farmer ID for the report: ")
farmer_report(entries, search_id)
Problems Output Ports Terminal
Debug Console
Filter (e.g. text, !exclude, \escape)
Terminal
Farmer Payment Report
Farmer ID: F001
Number of entries: 1
Total litres: 10
Total payment: ₹ 300.00
PS C:\Users\Admin\OneDrive\Desktop\CSA08\CSA08-Assignment1>
Outline
Timeline
main*
s.m yuvashri (18 hours ago)
Ln 53, Col 34
Spaces: 4
UTF-8
CRLF
Python
Python 3.15 (64-bit)
29°C
Partly sunny
Search
ENG
IN
13:57
24-10-2026


File Edit Selection View Go Run
CSA08-Assignment1
Sign In
Explorer
CSA08-Assignment1
milk_records.csv
README.md
Task01_analysis.md
Task02_Python_Basic...
Task03_Milk_Rate.py
Task04_Payment.py
Task05_Validation.py
Task06_Multiple_Entr...
Task07_Ten_Day_Tota...
Task08_Shift_Co... M
Task09_Farmer_R... M
Task10_Highest_Pay...
Task11_Save_Record...
Task12_Seasonal_Cal...
Outline
Timeline
Untitled-3
Task07_Ten_Day_Total.py
Task08_Shift_Count.py M
Task09_Farmer_Report.py M
Task10_Highest_Payment.py
Task10_Highest_Payment.py > ...
28
29 if litres <= 0 or rate < 0:
30 print("Invalid quantity or rate. Entry skipped.")
31 continue
32
33 payment = (litres * rate).quantize(Decimal("0.01"))
34
35 entries.append({
36 "farmer_id": farmer_id,
37 "litres": litres,
38 "payment": payment
39 })
40
41 except Exception:
42 print("Invalid input. Entry skipped.")
43
44 find_highest_payment(entries)
Problems Output Ports Terminal
Debug Console
Filter (e.g. text, !exclude, \escape)
Terminal
Highest Payment Record
Farmer ID: F002
Milk quantity: 20 litres
Payment: ₹ 600.00
PS C:\Users\Admin\OneDrive\Desktop\CSA08\CSA08-Assignment1>
power...
power...
power...
powersh...
powersh...
powersh...
Python De...
main* 0 0
s.m yuvashri (18 hours ago)
Ln 39, Col 11
Spaces: 4
UTF-8
CRLF
Python
Python 3.15 (64-bit)
29°C
Partly sunny
Search
ENG
IN
13:58
24-10-2026


File Edit Selection View Go Run
CSA08-Assignment1
Sign In
Explorer
CSA08-Assignment1
milk_records.csv M
README.md
Task01_analysis.md
Task02_Python_Basic...
Task03_Milk_Rate.py
Task04_Payment.py
Task05_Validation.py
Task06_Multiple_Entr...
Task07_Ten_Day_Tota...
Task08_Shift_Co... M
Task09_Farmer_R... M
Task10_Highest_Pay...
Task11_Save_Record...
Task12_Seasonal_Cal...
Outline
Timeline
en_Day_Total.py
Task08_Shift_Count.py M
Task09_Farmer_Report.py M
Task10_Highest_Payment.py
Task11_Save_Records.py
Task11_Save_Records.py > ...
40 snf_change = (snf - Decimal("8.5")) / Decimal("0.1")
41
42 rate = (
43 Decimal("30.00")
44 + fat_change * Decimal("0.40")
45 + snf_change * Decimal("0.30")
46 ).quantize(Decimal("0.01"), rounding=ROUND_HALF_UP)
47
48 payment = (litres * rate).quantize(
49 Decimal("0.01"),
50 rounding=ROUND_HALF_UP
51 )
52
53 save_record(farmer_id, litres, fat, snf, shift, rate, payment)
54
55 print("Rate per litre: ₹", rate)
56 print("Payment: ₹", payment)
Problems Output Ports Terminal
Debug Console
Filter (e.g. text, !exclude, \escape)
Terminal
Enter SNF percentage: 8.5
Enter shift (Morning/Evening): Morning
Rate per litre: ₹ 30.00
Payment: ₹ 300.00
Record saved to milk_records.csv
PS C:\Users\Admin\OneDrive\Desktop\CSA08\CSA08-Assignment1>
power...
power...
power...
powersh...
powersh...
powersh...
Python De...
main*
Ln 61, Col 1 Spaces: 4 UTF-8 CRLF { } Python Python 3.15 (64-bit)
29°C
Partly sunny
Search
ENG
IN
13:59
24-10-2026


File Edit Selection View Go Run
CSA08-Assignment1
Sign In
Explorer
CSA08-Assignment1
milk_records.csv M
README.md
Task01_analysis.md
Task02_Python_Basic...
Task03_Milk_Rate.py
Task04_Payment.py
Task05_Validation.py
Task06_Multiple_Entr...
Task07_Ten_Day_Tota...
Task08_Shift_Co... M
Task09_Farmer_R... M
Task10_Highest_Pay...
Task11_Save_Record...
Task12_Seasonal_Cal...
Outline
Timeline
ount.py M
Task09_Farmer_Report.py M
Task10_Highest_Payment.py
Task11_Save_Records.py
Task12_Seasonal_Calculator.py
Task12_Seasonal_Calculator.py > ...
rate = calculate_rate(fat, snf, month)
payment = litres * rate
bonus = Decimal("0.00")
if month in (4, 5, 6) and ten_day_litres > 500:
bonus = litres * Decimal("0.50")
total = (payment + bonus).quantize(
Decimal("0.01"),
rounding=ROUND_HALF_UP
)
print("\n--- Milk Payment Summary ---")
print("Farmer ID:", farmer_id)
print("Rate per litre: ₹", rate)
print("Milk payment: ₹", payment.quantize(Decimal("0.01")))
print("Seasonal bonus: ₹", bonus.quantize(Decimal("0.01")))
Problems Output Ports Terminal
Debug Console
Filter (e.g. text, !exclude, \escape)
Terminal
Farmer ID: F001
Rate per litre: ₹ 32.00
Milk payment: ₹ 640.00
Seasonal bonus: ₹ 10.00
Total payment: ₹ 650.00
PS C:\Users\Admin\OneDrive\Desktop\CSA08\CSA08-Assignment1>
power...
power...
power...
powersh...
powersh...
powersh...
Python De...
main*
Ln 75, Col 1
Spaces: 4
UTF-8
CRLF
Python
Python 3.15 (64-bit)
29°C
Partly sunny
Search
ENG
IN
14:00
24-10-2026